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Parenting Styles and Digital Game Addiction in Early Childhood

أنماط التربية وإدمان الألعاب الالكترونية في مرحلة الطفولة

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Abstract:

The increasing prevalence of digital game use among young children has raised concerns about early-onset behavioral addiction and its long-term developmental consequences. This paper provides a theoretical examination of how maternal parenting styles contribute to the formation of compulsive gaming habits in childhood. By integrating insights from developmental psychology and behavioral science, the paper explores the psychological mechanisms through which parenting influences a child's relationship with digital media—particularly in terms of emotional security, self-regulation, and behavioral modeling. It also considers how early exposure to screens, often used as a means of distraction or pacification, may condition children to rely on digital environments to fulfill unmet emotional needs. The paper proposes a conceptual framework that connects parenting patterns with digital dependency and outlines key strategies for early prevention, emphasizing balanced parenting and healthy play alternatives.

Keywords: Digital Game Addiction; Parenting Styles; Childhood; Attachment; Behavioral Addiction.

المخلص:

أصبح الإدمان على الألعاب الرقمية لدى الأطفال ظاهرة متنامية تستدعي قراءة نفسية وتربوية معمقة، خصوصاً في ظل شيوع ممارسات أسرية تُشجّع على الاستخدام المبكر والمفرط للأجهزة الذكية. تتناول هذه الورقة النظرية أثر أنماط تربية الأم في تشكيل سلوك الطفل الرقمي، بالاستناد إلى أطر تفسيرية متعددة تشمل نظرية التعلّق والنظرية المعرفية السلوكية. كما تُبرز الورقة كيف يُمكن لممارسات التهذئة الرقمية أن تخلق علاقة اعتمادية تُمهد للإدمان. ويُعرض نموذج مفاهيمي يربط بين أنماط التربية، والمتغيرات النفسية الوسيطة، وسلوكيات الإدمان، مع التوصية بتدخلات وقائية تُراعي حاجات الطفل النمائية والتفاعلية.

الكلمات المفتاحية: إدمان الألعاب الإلكترونية، أنماط التربية، الطفولة، التعلّق، الإدمان السلوكي.

1. INTRODUCTION

In light of the dramatic expansion of the contemporary digital environment, electronic games have become a central component of children's daily activities across the globe. Once limited to recreational purposes, these games have evolved into influential factors shaping children's psychological, social, and behavioral interactions (Spada, 2014; Elhai, Rozgonjuk, & Brailovskaia, 2021). While digital games can offer opportunities for cognitive development and learning, excessive use has been associated with the emergence of a compulsive behavioral pattern known as "Internet Gaming Disorder." This condition is characterized by loss of control, prioritization of gaming over essential life activities, and persistence despite negative consequences (American Psychiatric Association, 2013; World Health

Organization, 2018).

Childhood is among the most sensitive developmental stages, during which interactive and physical play contributes fundamentally to the formation of motor, emotional, and social competencies. It serves as a primary medium for building identity and fostering creativity (Malik & Marwaha, 2018; Plass, Homer, & Kinzer, 2015). However, the shift from free, physical play to digital gaming platforms has disrupted this developmental balance. Children now fulfill their innate need for play through closed digital environments that lack meaningful human interaction, reinforcing social isolation and limiting natural opportunities for socio-emotional learning (Ardoin & Bowers, 2020).

In recent years, a growing trend has been observed among mothers who routinely hand

over smartphones to their children at a very early age—sometimes as young as 6 to 8 months—as a way to calm them down or prevent disruptive behavior. This behavior is commonly seen at home, in waiting rooms, on public transport, and even during family gatherings. The screen becomes a substitute for human engagement—an indirect form of "educational distraction." Although seemingly harmless, this practice reflects a deeper misalignment in foundational parenting practices and may foster early dependency on digital devices, paving the way for future behavioral disorders such as digital addiction (Chou et al., 2005; Ryan & Deci, 2017).

While prior research has explored the clinical characteristics and consequences of gaming addiction, there remains a noticeable gap in understanding how early, everyday parenting behaviors—particularly the use of screens as pacifiers—might contribute to the psychological and environmental predisposition toward digital dependency. Moreover, few studies have examined the relationship between digital game addiction and the child's unmet developmental need for real, interactive play, or the regulating role of early parental care in moderating the child's digital experience.

This paper aims to offer a theoretical and analytical exploration of digital game addiction in children. It situates the phenomenon within a developmental psychology framework, examining the essential role of play in child development and employing contemporary psychological theories to explain digital addiction as a function of parenting patterns—specifically maternal styles. The article also proposes preventive and intervention strategies tailored to this critical developmental period.

2. Recent Empirical Studies on Digital Game Addiction in Childhood

In order to contextualize the current theoretical analysis, this section presents an overview of recent empirical studies that have explored the relationship between parenting styles and digital game addiction among children. The selected studies reflect

diverse methodological approaches and geographical contexts, providing a comparative foundation for understanding how parenting factors influence children's digital behavior.

Author(s) & Year	Country / Sample	Key Variables	Measurement Tools	Main Findings	Study Contribution
Gunawan et al. (2023)	Indonesia / 412 children aged 8–12	Digital attachment, gaming addiction, parental bond	Digital Attachment Scale, IAT	Strong digital attachment was significantly mediated by emotional detachment from caregivers	Highlighted the protective role of warm parenting
Lestari et al. (2023)	Indonesia / Parents & children	Parental involvement, screen time	Parent Survey, Screen Time Diary	Parent-based interventions reduced problematic screen use by 27%	Demonstrated the effectiveness of community parental training
Lu et al. (2023)	Systematic Review (Asia & Europe)	Gaming disorder, CBT-based prevention	Meta-analysis	CBT-based programs (e.g., PROTECT) reduced symptoms in school-aged children	Emphasized preventive intervention strategies
Tóth-Király et al. (2020)	Hungary / 529 adolescents	Psychological needs, gaming behavior	Basic Psychological Needs Scale, IGD-20	Frustration of psychological needs predicted gaming addiction	Applied Self-Determination Theory empirically
Nikken & Scholtes (2015)	Netherlands / 209 families	Parenting styles, screen regulation	Parental Mediation Questionnaire	Permissive parenting linked to excessive digital play	Connected parenting style with child's self-regulation capacities

Table 1. Comparative Overview of Recent Empirical Studies on Digital Game Addiction and Parenting Factors in Childhood

The comparative overview presented in **Table 1** reveals a growing scholarly consensus regarding the multifactorial nature of digital game addiction in children. Notably, studies underscore the critical role of parenting—particularly emotional availability and regulatory practices—in shaping children's relationship with digital media. Research such as that by Gunawan et al. (2023) and Nikken & Schols (2015) emphasizes how permissive or emotionally distant parenting can foster early dependency, while intervention-based studies (e.g., Lestari et al., 2023; Lu et al., 2023) demonstrate the effectiveness of structured parental involvement in mitigating problematic screen use. These findings align closely with the theoretical assumptions of the present article and provide a solid empirical basis for the proposed psychological model of early digital addiction.

3. Theoretical Background and Core Concepts

Understanding digital game addiction in children requires first clarifying the concept of behavioral addiction, its clinical features, the role of digital versus physical play, and the developmental importance of play in early childhood:

3.1 .Behavioral Addiction: Definition and Clinical Framework

To understand the phenomenon of digital game addiction in childhood, it is essential to first clarify the broader conceptual framework of behavioral addiction. Unlike substance-related addictions, behavioral addictions involve the compulsive engagement in certain activities—such as gambling, shopping, or digital gaming—despite significant negative consequences to an individual's psychological, social, or physical well-being (Goodman, 1990; Grant et al., 2010). The reinforcement in these cases is psychological rather than chemical, derived from the activity itself and its effects on emotional regulation and reward processing.

Digital gaming, specifically, has emerged in recent years as one of the most prominent forms of behavioral addiction, particularly

among children and adolescents. The immersion, achievement systems, and interactive feedback loops found in many games make them especially potent reinforcers, triggering similar reward mechanisms in the brain to those activated by addictive substances.

3.2 .Digital Game Addiction in Children: Characteristics and Diagnostic Indicators

The term “digital game addiction” refers to a pattern of excessive and compulsive gaming behavior characterized by a loss of control over playtime, neglect of other life domains (social, academic, physical), and continued use despite adverse consequences (Petry et al., 2015). The World Health Organization has recognized this condition under the classification of “Gaming Disorder” in the ICD-11, designating it as a diagnosable behavioral disorder (WHO, 2018).

According to the American Psychiatric Association (2013), diagnostic criteria for Internet Gaming Disorder include persistent preoccupation with gaming, withdrawal symptoms when not playing, loss of interest in other activities, and the continuation of play despite awareness of its negative effects—persisting for at least 12 months. The early onset of such patterns during childhood is particularly concerning, as it may establish dependency behaviors that are difficult to reverse later in life.

3.3 .The Difference Between Physical and Digital Play

Developmental psychology emphasizes the essential role of real-world play in fostering children's cognitive, emotional, and social growth. Through physical and imaginative play, children learn to explore their environment, build motor skills, engage in social negotiation, and express themselves in symbolic and experiential ways (Isaacs, 2018; Malik & Marwaha, 2018). This type of play also promotes creativity, emotional resilience, and interpersonal sensitivity.

In contrast, digital play—while offering certain forms of stimulation—typically lacks the depth of human interaction, physical engagement, and spontaneous exploration. It operates within closed systems with pre-

programmed responses and goals, which may limit the child's opportunity to develop real-world coping skills and emotional flexibility (Plass et al., 2015; Ardoin & Bowers, 2020).

3.4 .Play as a Developmental Need in Childhood

Play is not merely recreational; it is a fundamental developmental need. It serves as a critical channel for identity formation, autonomy, emotional expression, and social integration. When this need is unmet—or is substituted prematurely with digital alternatives—it may result in psychological imbalance and unmet developmental tasks. This, in turn, increases vulnerability to maladaptive behaviors, including digital addiction.

Recent research links digital overuse with the frustration of basic psychological needs such as competence, relatedness, and autonomy (Tóth-Király et al., 2020; Ryan & Deci, 2017). Consequently, digital game addiction should be conceptualized not simply as a behavioral excess but as a reflection of deeper disturbances in the child's developmental ecosystem.

4. Theoretical Frameworks Explaining Digital Game Addiction in Children

Understanding digital game addiction requires more than descriptive analysis—it calls for psychological models that clarify the underlying mechanisms. Several prominent theories offer valuable insights into how parenting patterns may contribute to early dependency on digital games.

4.1 .Self-Determination Theory (Deci & Ryan, 2000)

This theory posits that human behavior is driven by the need to fulfill three basic psychological needs: competence, relatedness, and autonomy. When these needs are not met—due to emotional neglect, rigid control, or lack of stimulation—children may turn to digital games as a substitute. Games provide structured goals, virtual social interaction, and a sense of mastery, thus satisfying these needs in an artificial and potentially addictive way (Ryan & Deci, 2017; Tóth-Király et al., 2020).

4.2 .Attachment Theory (Bowlby, 1969)

Secure attachment to caregivers fosters emotional regulation and healthy exploration. In contrast, insecure attachment—often resulting from inconsistent or emotionally unavailable parenting—can push children to seek comfort through external objects such as digital devices. These tools may become "digital attachment figures," providing the illusion of safety and predictability (Schimmenti et al., 2014).

4.3 .Cognitive Behavioral Theory (Beck, 1976; Young, 1998)

Children's engagement in gaming may stem from distorted thoughts, such as "Only games make me feel competent" or "The device is my only friend." These beliefs are reinforced by the immediate rewards and structured environments of games. Parenting practices that use devices as tools for reward or punishment can further entrench these maladaptive associations (King & Delfabbro, 2014).

4.4 .Social Learning Theory (Bandura, 1977)

Children learn behaviors by observing key role models, especially parents. When a child sees their mother frequently using her phone or using it as a pacifying tool, the child learns that emotional regulation is linked to screen use. This observed reinforcement contributes to the formation of habitual, compulsive screen behaviors (Gentile et al., 2012).

5. Psychological and Social Factors Contributing to Digital Game Addiction

Digital game addiction in childhood results from a complex interplay between individual traits, family dynamics, and broader social environments. Identifying these contributing factors is essential for understanding the roots of dependency and informing effective prevention strategies.

5.1 .Psychological Factors: Self-Regulation and Unmet Needs

Children with poor emotional self-regulation—such as difficulty managing impulses or delaying gratification—are more likely to engage in compulsive gaming. These difficulties often stem from underdeveloped executive functioning or lack of supportive parenting (Mazur et al.,

2022). Moreover, when basic psychological needs such as recognition, belonging, or security are unmet in real life, children may compensate through the immediate gratification provided by games (Tóth-Király et al., 2020).

5.2 .Family and Parenting Practices

Parent-child relationships are central to digital habits. Authoritarian or neglectful parenting can foster emotional gaps that lead children to overuse screens for comfort. In contrast, warm and structured parenting has been shown to protect against excessive use (Gentile et al., 2011). Studies also highlight the risks of permissive parenting, where limited boundaries allow uncontrolled digital engagement (Nikken & Schols, 2015; Rzymiski et al., 2023).

5.3 .Social and Environmental Contexts

Urban overcrowding, lack of safe outdoor spaces, and reduced social interaction have limited children's opportunities for real-life play. In this vacuum, digital games become the default leisure option. Additionally, when children observe widespread screen use among adults, they are more likely to imitate and internalize such behaviors as normative (Moudatsou et al., 2021).

5.4 .Design of Digital Games

Many games are intentionally engineered to maximize engagement using behavioral reinforcement techniques—such as rewards, levels, and notifications—that activate dopamine pathways. These mechanisms enhance the addictive potential of games, especially in young users with immature regulatory systems (Dong et al., 2019; Weinstein et al., 2022).

6. Psychological and Social Impacts of Digital Game Addiction in Children

Excessive engagement in digital games during childhood can have profound effects on mental health, behavior, and social development. While gaming may begin as a harmless activity, its overuse may disrupt the child's emotional balance, cognitive functioning, and interpersonal relationships.

6.1 .Psychological Effects: Mood Dysregulation and Emotional Issues

Studies have consistently linked problematic

gaming with symptoms of anxiety, depression, irritability, and emotional dysregulation. Addicted children often experience frustration, mood swings, and difficulty expressing emotions, largely due to reduced real-world social interaction and emotional learning (Maroney et al., 2019; Weinstein et al., 2022). Furthermore, games may create a false sense of competence, leading to lowered self-esteem when real-life expectations are not met (Lemmens et al., 2011).

6.2 .Behavioral and Academic Effects

Children who spend excessive time gaming often show attention problems, reduced academic motivation, and lower performance in subjects requiring sustained focus (Cheng et al., 2021). Disrupted sleep patterns—especially when gaming occurs late at night—can further impair learning and memory consolidation.

6.3 .Social Consequences: Withdrawal and Isolation

Addicted children may withdraw from real-life relationships and prefer virtual interactions, which impedes their ability to develop effective social skills and handle frustration. Over time, this can contribute to isolation, reduced empathy, and limited capacity for emotional reciprocity (Kuss & Griffiths, 2015).

These findings highlight the need for early awareness and structured intervention strategies, especially during critical developmental periods.

7. Parenting Styles and Their Relationship to Digital Game Addiction

Maternal parenting style plays a pivotal role in shaping a child's emotional regulation, self-discipline, and engagement with digital media. Diana Baumrind's (1971) classic typology identifies four main parenting styles, each with distinct implications for children's psychological development and risk of digital addiction.

7.1 .Authoritarian Parenting

Characterized by strict control and limited emotional expression, this style may lead children to seek autonomy and emotional relief in digital environments. Studies

suggest that such children are more prone to escapism and compulsive gaming (Yaffe, 2020).

7.2 .Authoritative Parenting

This balanced approach—combining warmth with clear boundaries—has been associated with healthier digital habits. Children raised under authoritative parenting tend to have stronger self-regulation skills and are less likely to develop addictive behaviors (Padilla-Walker et al., 2012).

7.3 .Permissive Parenting

Marked by high acceptance but low discipline, this style often results in fewer behavioral boundaries. Children in such environments may engage excessively with digital games due to the absence of structure (Nikken & Schols, 2015).

7.4 .Neglectful Parenting

In environments with emotional detachment and minimal supervision, children may turn to digital devices to fill emotional voids. These children are considered highly vulnerable to digital dependency (Rzymiski et al., 2023).

Overall, the type of maternal care a child receives can either buffer or amplify the psychological conditions that lead to digital game addiction.

8. Appropriate Age for Smartphone Use: Balancing Access and Risk

The question of when to introduce children to smartphones is a growing concern in developmental and digital health research. While exposure to technology is often seen as inevitable, the timing and manner of its introduction have critical implications for cognitive and emotional development.

8.1. Expert Guidelines and Developmental Risks

The American Academy of Pediatrics (AAP, 2016) recommends avoiding screen exposure entirely for children under 18 months, except for video calls. For toddlers aged 18–24 months, only high-quality educational content should be allowed—and only with active parental engagement. For children over two years, screen time should be limited to one hour per day, ideally co-watched with a caregiver.

8.2. Real-World Trends and Associated Concerns

Despite these recommendations, many mothers expose children to smartphones as early as 6–8 months to distract or soothe them. While convenient, this practice can disrupt the development of executive functioning, attention regulation, and emotional coping (Christakis, 2019). Early and frequent screen exposure has also been linked to speech delays, sleep disturbances, and symptoms of ADHD (Madigan et al., 2019).

The earlier the introduction of digital devices, the higher the risk of dependency, particularly if usage is unsupervised and emotionally compensatory.

.9. Prevention Strategies for Digital Game Addiction in Children

Given the multifaceted nature of digital addiction, effective prevention must involve coordinated efforts across family, school, and community systems. Recent research emphasizes interventions that are proactive, developmentally appropriate, and context-sensitive.

9.1 .Focus on Usage Patterns, Not Just Screen Time

Recent evidence shows that compulsive behavior (e.g., irritability, withdrawal) is a stronger predictor of psychological distress than the total time spent on screens (Nagata et al., 2022). Monitoring how children use technology is more important than simply counting hours.

9.2 .Parent-Based Interventions

Strong parental involvement reduces the risk of device dependency. Studies indicate that parenting programs focusing on communication and emotional support help buffer the effects of digital overuse (Gunawan et al., 2023; Lestari et al., 2023).

9.3 .Cognitive-Behavioral Prevention (CBT)

CBT-based programs, such as the PROTECT intervention, have proven effective in reducing gaming disorder symptoms among school-age children (Lu et al., 2023). These programs focus on cognitive reframing, behavioral regulation, and digital literacy.

9.4 .Digital Tools for Shared Engagement

Smart systems and co-playing apps (e.g., ShieldUp!) encourage collaborative screen use between parents and children, reducing

solitary and compulsive gaming behaviors (Gunawan et al., 2023).

9.5 .School and Community Programs

Educational programs that combine family and school efforts enhance children's digital self-regulation. Activities such as sports, art, and social clubs serve as healthy offline alternatives (Lestari et al., 2023).

Prevention must start early, with a strong foundation in emotional bonding, digital literacy, and structured guidance—especially in the formative years of development.

10. Recommendations

Based on the theoretical and empirical analysis presented in this paper, the following recommendations are proposed:

1. Delay Digital Exposure: Avoid introducing smartphones and gaming devices before the age of two. After this age, use should be structured, limited, and accompanied by parental guidance.

2. Parental Awareness Campaigns: Design psychoeducational programs that help parents understand the developmental risks of excessive screen exposure and the importance of play-based learning.

3. Strengthen Emotional Bonds: Encourage secure attachment between caregivers and children through responsive and emotionally attuned parenting, as a protective factor against digital dependency.

4. Integrate Prevention in Schools: Develop school-based interventions focused on emotional regulation, attention training, and offline recreational engagement.

5. Regulate Game Design: Advocate for ethical game development that avoids manipulative reinforcement systems, especially in games marketed to children.

6. Encourage Alternative Activities: Promote participation in sports, arts, and real-world social interaction to satisfy children's need for play, achievement, and belonging in a healthy way.

11. Conclusion

Digital game addiction during childhood should not be interpreted solely as a symptom of excessive screen exposure, but rather as a manifestation of broader psychological and environmental

dynamics—especially those shaped by early parenting patterns. As this paper has demonstrated, the maternal style of caregiving plays a central role in determining how children relate to the digital world. Emotional neglect, permissive or authoritarian parenting, and the absence of structured play opportunities may all contribute to a child's vulnerability to compulsive gaming.

Drawing on several psychological theories—such as Self-Determination Theory, Attachment Theory, Cognitive-Behavioral Theory, and Social Learning Theory—this article has shown that unmet emotional needs, poor self-regulation, and insecure parent-child relationships serve as key mediators between parenting and addictive behavior. When the foundational need for connection, competence, and autonomy is not satisfied through healthy relationships and real-world play, children may seek digital games as a substitute—one that is immediately rewarding but ultimately developmentally limiting.

The implications of these findings are both theoretical and practical. Theoretically, they call for a more integrative understanding of behavioral addictions in childhood—one that incorporates family dynamics, emotional development, and environmental context. Practically, they highlight the urgency of early interventions that focus not only on limiting screen time but also on promoting secure attachment, parental digital literacy, and structured offline engagement.

In an era where digital devices are omnipresent, and often used as tools of pacification or distraction, it becomes imperative to rethink the emotional and educational functions of technology in childhood. Smartphones and games cannot replace the developmental richness of human interaction, nor should they become substitutes for parental presence.

Future research should examine longitudinal trajectories of digital game addiction, especially in relation to specific parenting styles, cultural contexts, and neurodevelopmental factors. Such research

will help refine prevention strategies and inform public policy aimed at fostering digital well-being from early childhood.

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